Preview: y2007_q03 (v1)

Comparing sets of letters: each different letter counts once; check them all

Part A: Each question is worth 3 points

1. How many letters do the words ELEPHANT and NUMBER have in common?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

2. How many letters do the words ELEPHANT and SPIDER have in common?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

3. How many different letters are used to write the word CHOCOLATE?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

4. How many different letters are used to write the word WINDOW?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

Part B: Each question is worth 4 points

5. Which letters of the word MATHEMATICS do *not* appear in the word TRIANGLE? How many different letters are there like that?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 8

6. Which letters of the word BANANA do *not* appear in the word SQUARE? How many different letters are there like that?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

7. Which word has exactly 2 different letters in common with DRAGON?

A. ROCKET ***B.*** TRIANGLE ***C.*** CASTLE ***D.*** PUZZLE ***E.*** LEMONADE

8. Which word has exactly 3 different letters in common with TEACHER?

A. ROCKET ***B.*** KANGAROO ***C.*** CASTLE ***D.*** GARDEN ***E.*** TRIANGLE

Part C: Each question is worth 5 points

9. Mia has letter cards and wants to show which letters are used in the words MATHEMATICS and PLANET. Mia puts down one card for each different letter used in either word. How many cards does Mia put down?

A. 17 ***B.*** 14 ***C.*** 12 ***D.*** 11 ***E.*** 10

10. Mia has letter cards and wants to show which letters are used in the words CHOCOLATE and RAINBOW. Mia puts down one card for each different letter used in either word.
How many cards does Mia put down?

A. 12 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 8

Answer Key

1.	**C**	**2.**	**E**	**3.**	**B**	**4.**	**A**	**5.**	**C**	**6.**	**D**
7.	**A**	**8.**	**D**	**9.**	**D**	**10.**	**A**				

Solutions

1. C (2) — Check each different letter of ELEPHANT: only E and N are also in NUMBER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

2. E (2) — Check each different letter of ELEPHANT: only E and P are also in SPIDER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

3. B (7) — The letters are A, C, E, H, L, O and T: 7 different letters (CHOCOLATE has 9 letters in all). [2007 Q3 · different_letters]

4. A (5) — The letters are D, I, N, O and W: 5 different letters (WINDOW has 6 letters in all). [2007 Q3 · different_letters]

5. C (4) — MATHEMATICS uses 8 different letters; A, E, I and T are also in TRIANGLE. The others, C, H, M and S, make 4. [2007 Q3 · only_in_first]

6. D (2) — BANANA uses 3 different letters; A are also in SQUARE. The others, B and N, make 2. [2007 Q3 · only_in_first]

7. A (ROCKET) — ROCKET shares O and R with DRAGON. PUZZLE shares 0; CASTLE shares 1; LEMONADE shares 4; TRIANGLE shares 4. [2007 Q3 · which_word]

8. D (GARDEN) — GARDEN shares A, E and R with TEACHER. ROCKET shares 4; TRIANGLE shares 4; CASTLE shares 4; KANGAROO shares 2. [2007 Q3 · which_word]

9. D (11) — MATHEMATICS uses 8 different letters and PLANET uses 6. A, E and T are in both, so count them once: 8 + 6 – 3 = 11. [2007 Q3 · letter_cards]

10. A (12) — CHOCOLATE uses 7 different letters and RAINBOW uses 7. A and O are in both, so count them once: 7 + 7 – 2 = 12. [2007 Q3 · letter_cards]